

Week 1

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pranav_chowdary@Pranav:~$ nano q1.c
pranav_chowdary@Pranav:~$ gcc q1.c
pranav_chowdary@Pranav:~$ ./a.out
Enter Linux command: ls
Parent PID: 617
Child PID: 618
a.out main.c q1.c
Child completed
```



```
GNU nano 8.7.1 q1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

```
pranav_chowdary@Pranav:~$ uname -a
Linux Pranav 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
pranav_chowdary@Pranav:~$ lscpu
Architecture:            x86_64
  CPU op-mode(s):        32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                  20
  On-line CPU(s) list:   0-19
Vendor ID:               GenuineIntel
Model name:              13th Gen Intel(R) Core(TM) i7-13645HX
CPU family:              6
Model:                   183
Thread(s) per core:      2
Core(s) per socket:      10
Socket(s):               1
Stepping:                 1
BogoMIPS:                 5606.39
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse
                        2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_t
                        sc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt
                        tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb
                        stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms inv
                        pcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip wa
                        itpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_ca
                        pabilities

Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    480 KiB (10 instances)
  L1i:                    320 KiB (10 instances)
  L2:                     20 MiB (10 instances)
  L3:                     24 MiB (1 instance)
```

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NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-19
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:         Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
pranav_chowdary@Pranav:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
pranav_chowdary@Pranav:~$ ps
  PID TTY          TIME CMD
  401 pts/0    00:00:00 bash
  693 pts/0    00:00:00 ps
pranav_chowdary@Pranav:~$ top

```

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pranav_chowdary@Pranav:~$ top
top - 08:40:38 up 2 min, 1 user, load average: 0.06, 0.05, 0.02
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7792.7 total, 6682.6 free, 583.0 used, 680.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7209.7 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
  93 root        20   0   35460 12456 9260 S  0.7   0.2   0:00.42 systemd-udevd
 709 root        20   0   35464 6576 3348 S  0.3   0.1   0:00.01 (udev-worker)
    1 root        20   0   24556 15524 11612 S  0.0   0.2   0:01.38 systemd
    2 root        20   0   3180  2208 2072 S  0.0   0.0   0:00.10 init-systemd(Ub
    6 root        20   0   3180  2044 1940 S  0.0   0.0   0:00.00 init
  49 root        19  -1  42196 16344 15080 S  0.0   0.2   0:00.26 systemd-journal
  89 systemd+    20   0   22540 14604 11988 S  0.0   0.2   0:00.06 systemd-resolve
 172 root        20   0   2888  1952 1820 S  0.0   0.0   0:00.05 chronyd-starter
 173 root        20   0   4472  3148 2900 S  0.0   0.0   0:00.00 cron
 174 message+    20   0   8688  5396 4728 S  0.0   0.1   0:00.04 dbus-daemon
 181 root        20   0   44092 29724 14584 S  0.0   0.4   0:00.20 networkd-dispat
 252 root        20   0   18428  9404 8168 S  0.0   0.1   0:00.08 systemd-logind
 254 root        20   0 1857652 15508 12552 S  0.0   0.2   0:00.09 wsl-pro-service
 263 syslog      20   0  220548  5920 4644 S  0.0   0.1   0:00.06 rsyslogd
 294 _chrony      20   0   24248 11808 7228 S  0.0   0.1   0:00.05 chronyd
 319 _chrony      20   0  12092  2420 1776 S  0.0   0.0   0:00.00 chronyd
 353 root        20   0 123456 32628 18024 S  0.0   0.4   0:00.17 unattended-upgr
 356 root        20   0   5492  2712 2524 S  0.0   0.0   0:00.01agetty
 399 root        20   0   3188  1112  980 S  0.0   0.0   0:00.00 SessionLeader
 400 root        20   0   3204  1252 1108 S  0.0   0.0   0:00.01 Relay(401)
 401 pranav_+    20   0   6268  5532 3864 S  0.0   0.1   0:00.06 bash
 402 root        20   0   8644  5516 4676 S  0.0   0.1   0:00.01 login
 445 pranav_+    20   0  22752 13344 10868 S  0.0   0.2   0:00.12 systemd
 449 pranav_+    20   0  22912 4092 2088 S  0.0   0.1   0:00.00 (sd-pam)
 477 pranav_+    20   0   6136  5416 3864 S  0.0   0.1   0:00.04 bash
 705 pranav_+    20   0   8160  5980 3888 R  0.0   0.1   0:00.00 top
 708 root        20   0   35464 6576 3348 S  0.0   0.1   0:00.00 (udev-worker)

```